Predicting Adverse Event Risk in a CDISC Clinical Trial Pipeline

CDISC Pilot Study, Xanomeline vs Placebo

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Contents

1 Introduction and Overview

This pipeline demonstrates a complete ADaM data workflow built using the pharmaverse R ecosystem, the open-source initiative co-maintained by Roche, GSK, Pfizer, and other major pharmaceutical companies to standardize clinical programming in R. Source data is drawn from the CDISC pilot SDTM dataset, which represents a Phase III clinical trial comparing Xanomeline (low and high dose) against Placebo in Alzheimer's disease patients.

The pipeline follows the same standards used in FDA regulatory submissions: CDISC SDTM as input, ADaM datasets as analysis-ready output, TLFs (Tables, Listings, Figures) as the deliverables, and a downstream predictive model demonstrating how structured clinical data feeds into ML-driven patient stratification.

Safety Population by Treatment Arm

Treatment Arm	N
Placebo	86
Xanomeline High Dose	84
Xanomeline Low Dose	84

Baseline Demographics, Safety Population

Treatment Arm	N	Mean Age	SD Age	Median Age	% Female
Placebo	86	75.2	8.6	76.0	61.6
Xanomeline High Dose	84	74.4	7.9	76.0	47.6
Xanomeline Low Dose	84	75.7	8.3	77.5	59.5

2 Data Sources

All source data comes from the `pharmaversesdtm` R package, which contains CDISC-compliant SDTM test datasets maintained by the pharmaverse community.

Domain	Description	Granularity
DM	Demographics	One row per patient
EX	Exposure (dosing records)	Many per patient
AE	Adverse Events	Many per patient
DS	Disposition	One row per patient

3 ADaM Datasets

3.1 ADSL: Subject Level Analysis Dataset

ADSL is the foundation of every ADaM package. It contains exactly one row per patient and captures baseline characteristics, treatment assignment, and population flags. Every downstream dataset joins back to ADSL using `USUBJID` as the key.

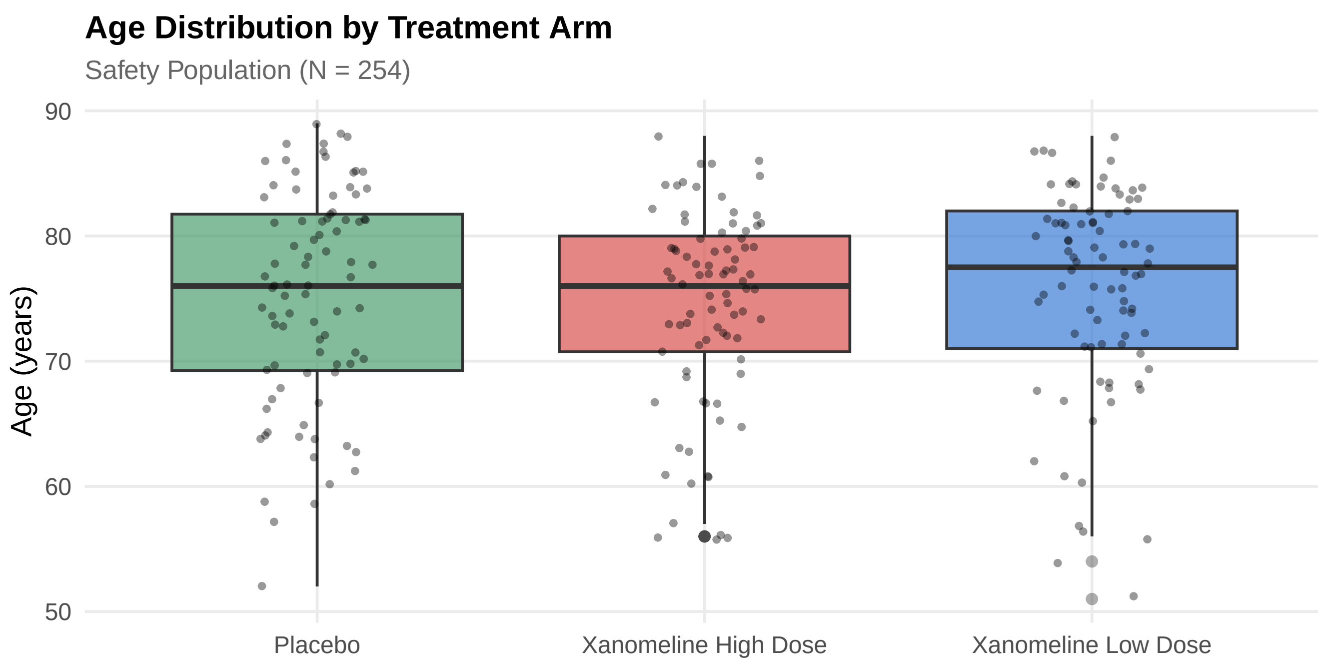
3.1.1 Population Summary

The trial enrolled 306 patients, of whom 254 formed the safety population. The remaining 52 patients were screen failures and are excluded from all downstream analyses.

Treatment-Emergent Adverse Events by Arm

Treatment Arm	Events	Patients	Events/Patient	% Serious
Placebo	281	65	4.32	0.0
Xanomeline High Dose	427	76	5.62	0.5
Xanomeline Low Dose	412	77	5.35	0.2

3.1.2 Baseline Demographics



The three arms are well balanced on age, with mean ages ranging from 74.4 to 75.7 years and overlapping distributions visible in the boxplot above. Sex distribution shows a notable imbalance, however: the Xanomeline High Dose arm enrolled 47.6% female versus 61.6% in Placebo and 59.5% in Low Dose. This warrants attention in any subsequent analysis comparing arms, since sex can confound tolerability comparisons.

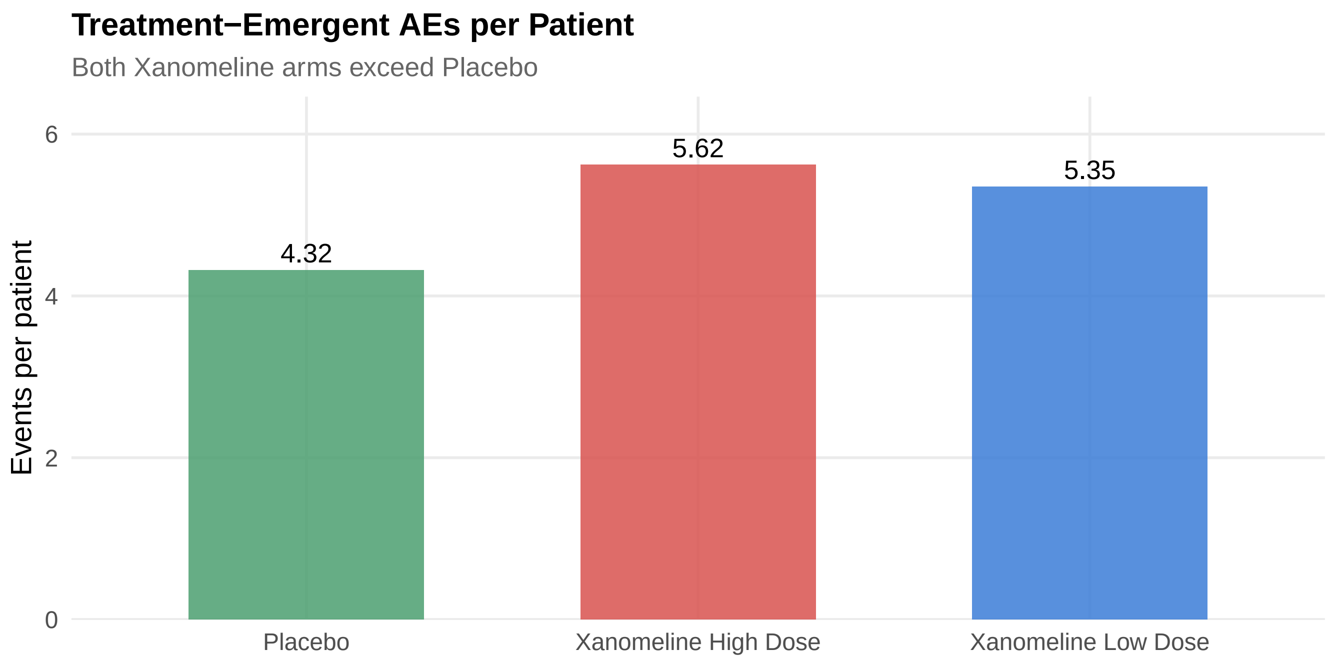
3.2 ADAE: Adverse Event Profile by Treatment Arm

The treatment-emergent flag (`TRTEMFL`) was derived per FDA guidance, distinguishing drug-related signals from pre-treatment conditions. This forms the basis for safety surveillance and patient stratification across treatment arms.

Top 10 System Organ Classes (TEAEs)

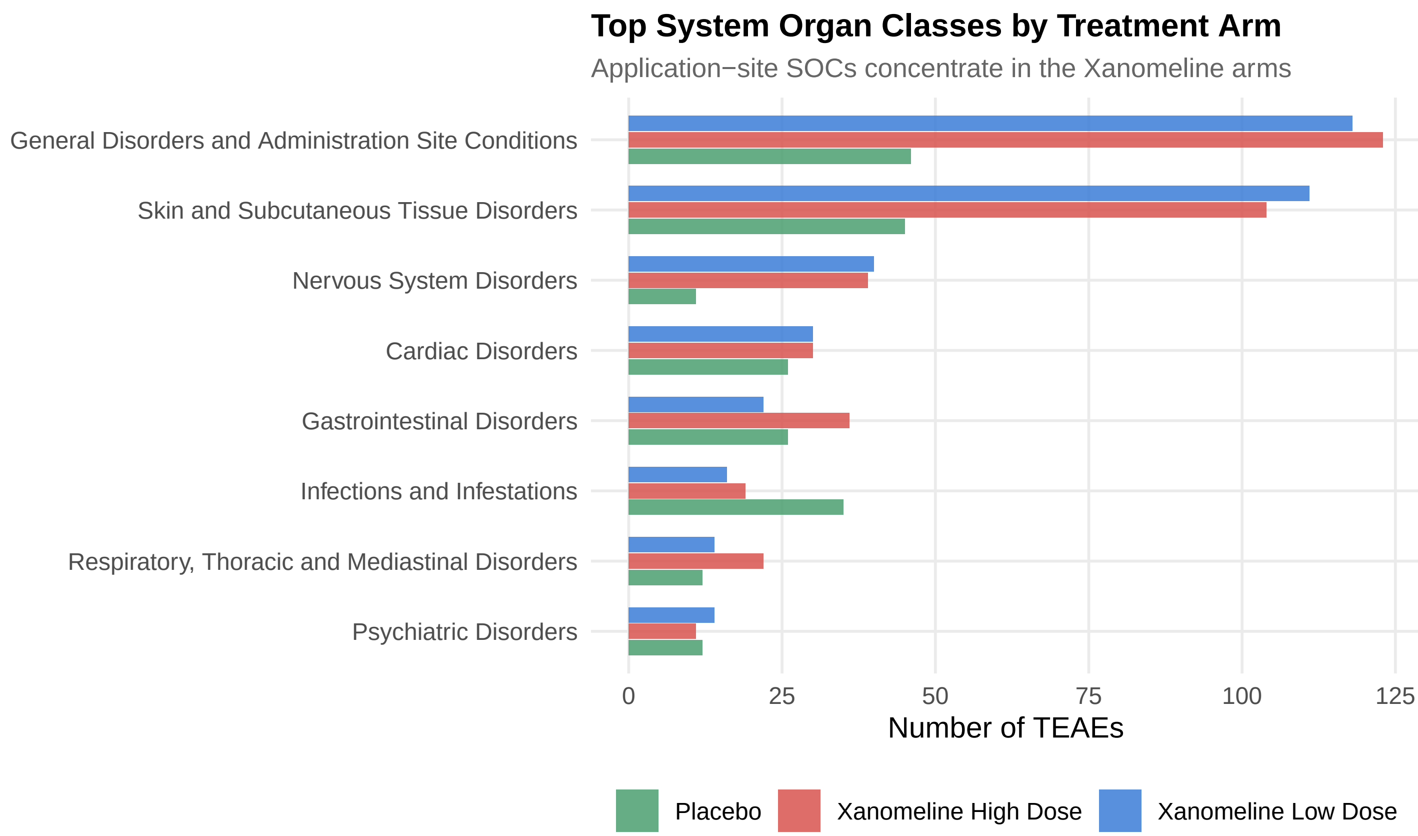
System Organ Class	Plac	Xan High	Xan Low	Total
GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS	46	123	118	287
SKIN AND SUBCUTANEOUS TISSUE DISORDERS	45	104	111	260
NERVOUS SYSTEM DISORDERS	11	39	40	90
CARDIAC DISORDERS	26	30	30	86
GASTROINTESTINAL DISORDERS	26	36	22	84
INFECTIONS AND INFESTATIONS	35	19	16	70
RESPIRATORY, THORACIC AND MEDIASTINAL DISORDERS	12	22	14	48
PSYCHIATRIC DISORDERS	12	11	14	37
INVESTIGATIONS	19	8	7	34
INJURY, POISONING AND PROCEDURAL COMPLICATIONS	9	8	12	29

3.2.1 Treatment-Emergent AE Overview



Both Xanomeline arms produced more events per patient than Placebo: 1.3x for High Dose (5.62 vs. 4.32 events/patient) and 1.24x for Low Dose (5.35 vs. 4.32). Serious AEs were rare across all arms (no greater than 0.5%), suggesting the differential signal is driven by frequency of nuisance events rather than severity.

3.2.2 Top System Organ Classes by Arm



The leading system organ class is **General Disorders and Administration Site Conditions**, with 123 events on Xanomeline High Dose and 118 on Low Dose, compared to 46 on Placebo. Together with skin and subcutaneous tissue disorders, these two SOCs disproportionately drive the Xanomeline safety signal, consistent with the known application-site tolerability profile of the Xanomeline transdermal patch. Systemic SOCs (cardiac, gastrointestinal, nervous system) appear comparable across arms.

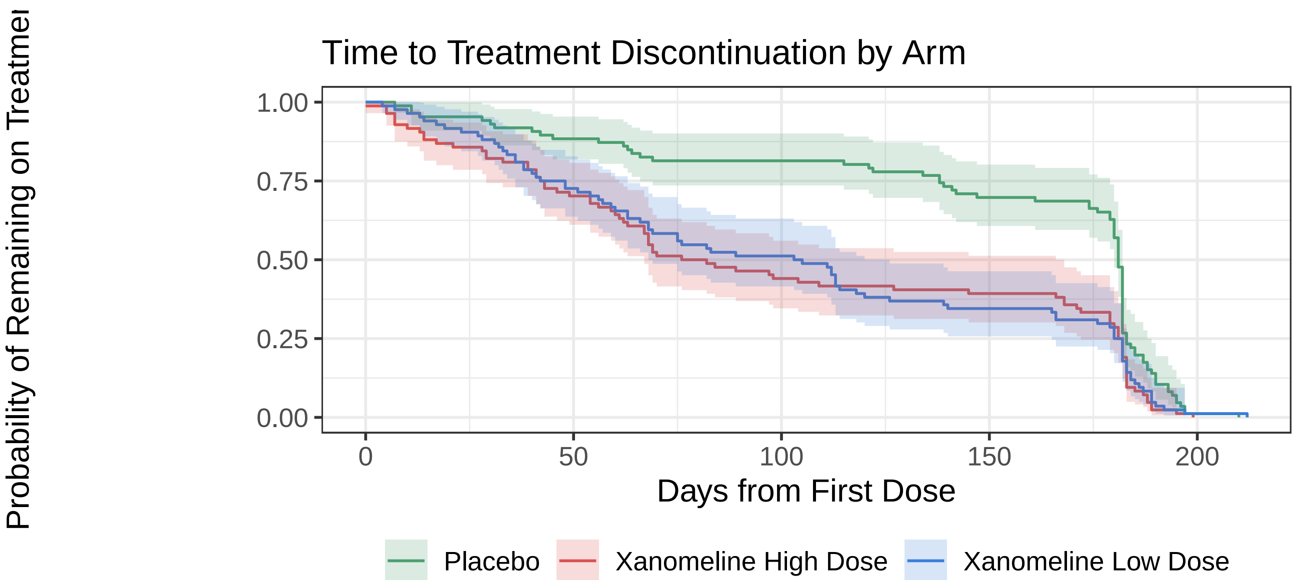
3.3 ADTTE: Time to Treatment Discontinuation

Kaplan-Meier analysis of time-to-discontinuation reveals differential tolerability across dose arms, directly informing dose selection and risk-benefit assessment.

Time to Discontinuation Summary

Treatment Arm	N	Events	Median Days
Placebo	86	86	181
Xanomeline High Dose	84	84	79
Xanomeline Low Dose	84	84	104

3.3.1 Kaplan-Meier Survival Curve



At Risk

	0	50	100	150	200
Placebo	86	76	70	60	1
Xanomeline High Dose	84	59	37	33	0
Xanomeline Low Dose	84	61	43	29	1

Events

	0	50	100	150	200
Placebo	0	10	16	26	85
Xanomeline High Dose	1	25	47	51	84
Xanomeline Low Dose	0	23	41	55	83

3.3.2 Log-Rank Test and Median Time-to-Event

The log-rank test is highly significant ($\chi^2 = 16.1$, 2 df, $p < 0.001$), confirming that time-to-discontinuation differs across arms. Median time on treatment was 181 days for Placebo, 104 days for Low Dose, and 79 days for High Dose. Patients on Placebo remained on treatment roughly 2.3x longer than those on High Dose, with a clear dose-response gradient between Low and High Dose arms.

4 Predictive Modeling: Baseline Risk Stratification

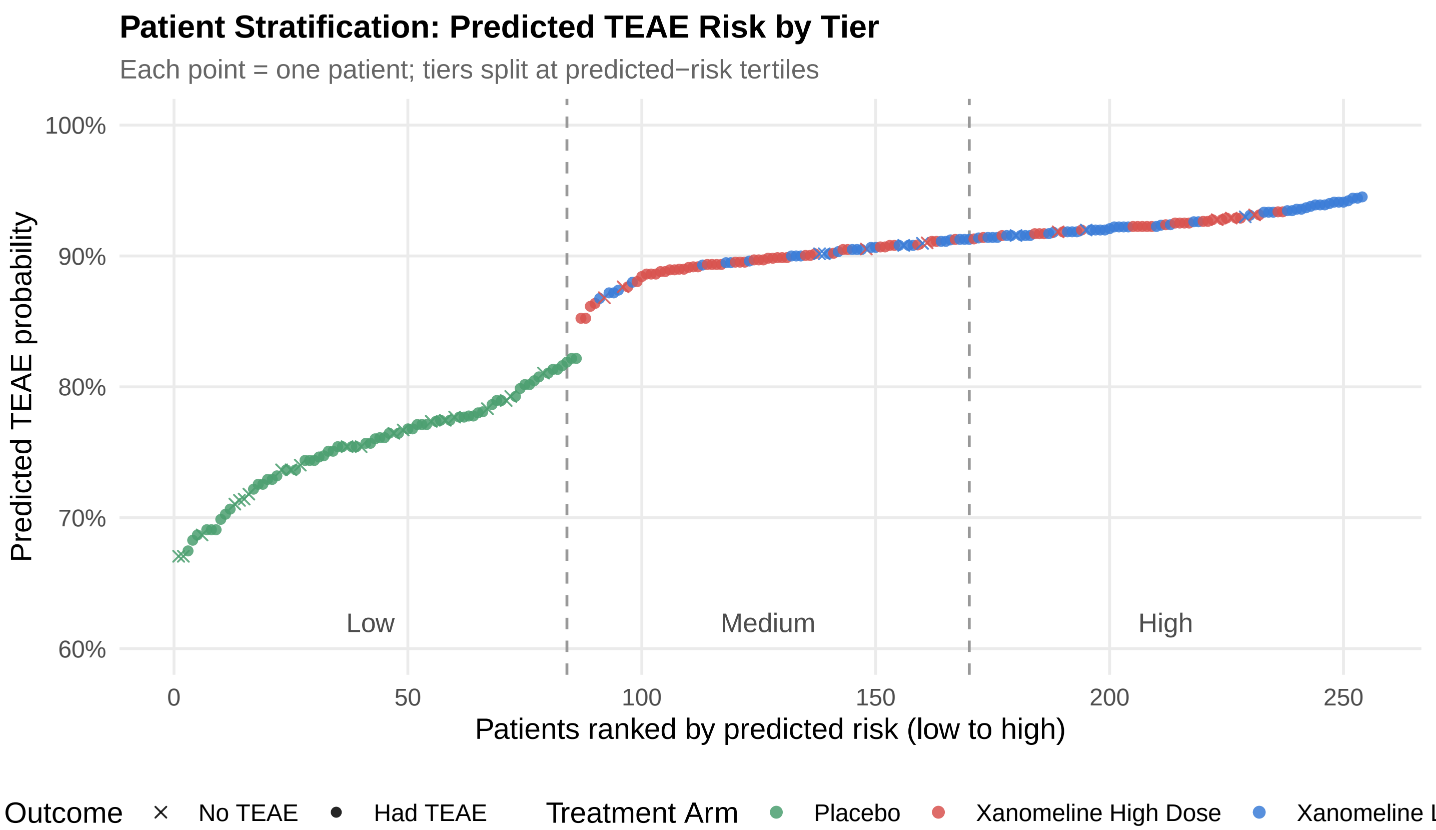
Patient stratification is the central goal of this section. Two complementary models are fit: a *baseline-only* risk model, using only pre-treatment characteristics (AGE, SEX), and a *full* model that adds treatment arm.

Keeping them separate matters because mixing them would conflate intrinsic patient risk with the trial's treatment effect; the baseline model captures who is at risk before any drug is given, while the full model layers on what the drug itself contributes.

The modeling cohort comprises **254 patients** in the safety population. The overall TEAE incidence is **85.8%**, an unusually high event rate that has implications for model performance discussed below.

4.1 Stratification Overview

Before walking through the modeling math, the chart below shows what stratification looks like in this cohort: every patient is plotted by their full-model predicted risk, sorted from lowest to highest, colored by treatment arm, and bucketed into Low/Medium/High tertiles. This is the visual the entire predictive section is building toward: a defensible way to sort patients by adverse event likelihood.



The structure here makes the dominant signal visible: Placebo patients (green) cluster in the Low tier, Xanomeline patients (blue and red) dominate Medium and High. Filled circles mark patients who actually had a TEAE; their density visibly increases left-to-right, which is what calibration is supposed to look like.

4.2 Model 1: Baseline-Only Risk

Baseline-only AUC: 0.569. Age and sex alone show essentially no signal for predicting TEAEs, consistent with the absence of strong univariate trends in this cohort. This is the appropriate "intrinsic patient risk" baseline before treatment is considered.

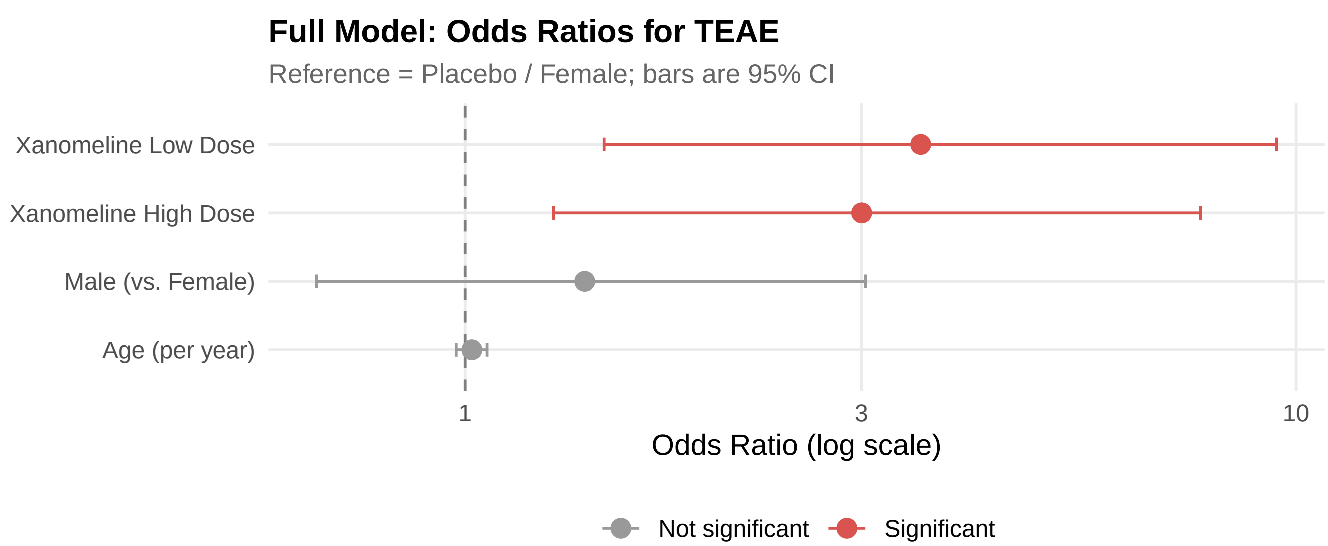
Baseline-Only Risk Factors (Odds Ratios)

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	1.387	1.612	0.203	0.8391	0.064	36.837
AGE	1.018	0.021	0.826	0.4089	0.975	1.061
SEXM	1.480	0.374	1.047	0.2953	0.720	3.159

Full Model Including Treatment Arm (Odds Ratios)

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	0.676	1.657	-0.236	0.8131	0.028	19.262
AGE	1.019	0.022	0.862	0.3889	0.976	1.063
SEXM	1.393	0.385	0.860	0.3896	0.662	3.033
ARMXanomeline High Dose	3.001	0.452	2.430	0.0151	1.278	7.677
ARMXanomeline Low Dose	3.534	0.469	2.692	0.0071	1.470	9.476

4.3 Model 2: Full Model with Treatment Arm



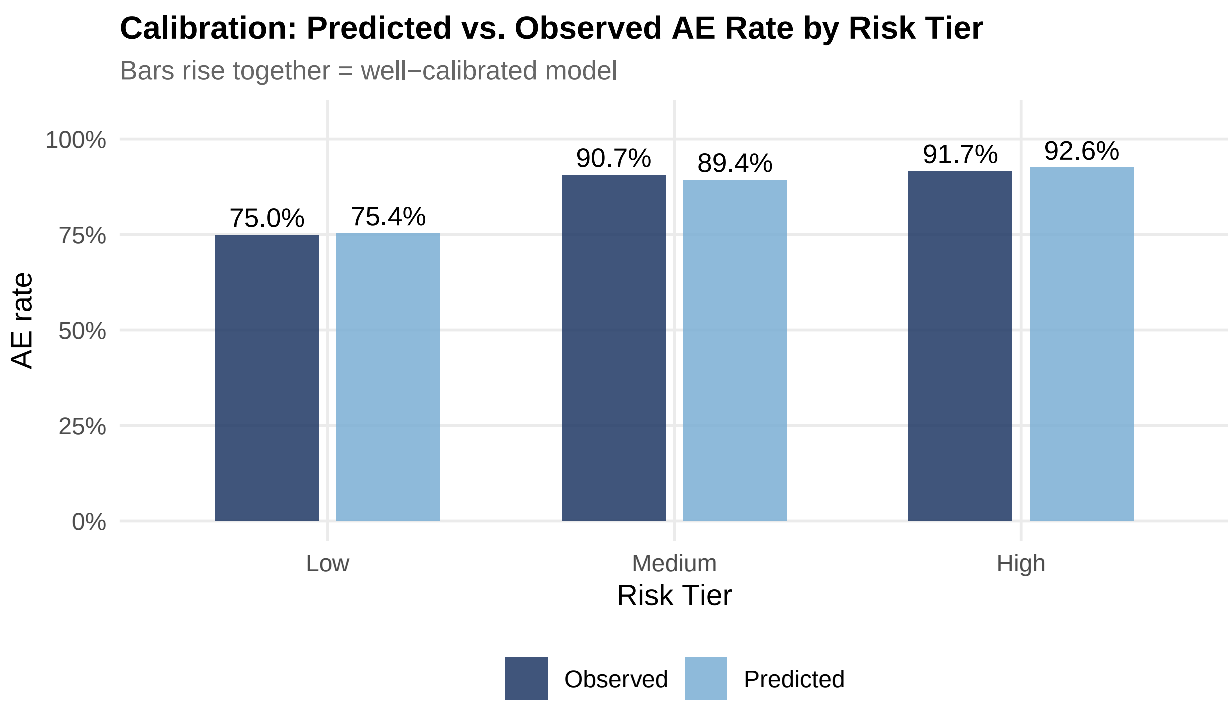
Treatment arm dominates the full model. Both Xanomeline arms carry significantly elevated TEAE odds vs. Placebo (Low Dose OR = 3.53, p = 0.0071; High Dose OR = 3, p = 0.015), while age and sex remain non-significant. The full-model AUC of **0.659** indicates modest discrimination: informative for population-level stratification but insufficient for confident individual-patient prediction.

4.4 Calibration Across Risk Tiers

Stratifying patients into low, medium, and high risk tertiles (based on the full-model predictions) and comparing predicted to observed AE rates demonstrates **calibration**, a critical property for any clinical decision-support model.

Calibration: Observed AE Rate by Predicted Risk Tier

Risk Tier	N	Mean Predicted Probability	Observed AE Rate
Low	84	0.754	0.750
Medium	86	0.894	0.907
High	84	0.926	0.917



Observed AE rates rise monotonically with predicted risk tier (75.0%, 90.7%, 91.7%), confirming the model is well-calibrated even though discriminative power is modest. This is the property that matters for decision support: a patient flagged "High risk" really does have a higher AE rate than one flagged "Low."

5 Limitations and Next Iteration

A rigorous report names the limits of its own analysis. Three are worth surfacing here.

Outcome saturation. With a 85.8% TEAE rate in the safety population, the binary "any TEAE" outcome is near-saturated, leaving little signal for discrimination. This explains why even a treatment-effect-laden model achieves only AUC 0.659. Three reformulations would extract more signal from the same data: (1) count of TEAEs per patient, modeled with negative binomial regression; (2) time to first TEAE, modeled with a Cox proportional hazards model; or (3) severe-only TEAEs as the binary outcome to focus on clinically meaningful events.

Limited baseline feature set. ADSL provides only AGE, SEX, and a small number of derived flags. In a real translational setting, the baseline-only model would be enriched with laboratory values (LB), prior medical history (MH), concomitant medications (CM), and biomarker measurements. The current AUC of 0.569 for AGE plus SEX is a floor, not a ceiling.

No held-out validation. All performance metrics are in-sample. A rigorous evaluation would split the cohort into training and validation sets, or use k-fold cross-validation, before reporting AUC and calibration.

6 Summary

This pipeline demonstrates the end-to-end workflow from CDISC SDTM source data through analysis-ready ADaM datasets to downstream decision tools. Source data was transformed into ADaM datasets using the pharmaverse R ecosystem, following the same standards used in FDA submissions, and adverse events were classified via the TRTEMFL flag to isolate 1,120 treatment-emergent events; the leading System Organ Classes pointed to application-site tolerability of the Xanomeline transdermal patch. A Kaplan-Meier analysis with log-rank testing ($p < 0.001$) showed a clear dose-response gradient in time-on-treatment, and two logistic regression models (one baseline-only, one with treatment arm) were fit to separate underlying patient risk from the treatment effect, with calibration holding across predicted risk tiers. The report also names its own limits: outcome saturation, a thin baseline feature set, and the absence of held-out validation as the natural next steps. The same architecture extends cleanly to other research settings where structured clinical or biological data needs to move from raw collection through analysis-ready form to a model that informs a real decision.